

# SMART PREDICTION OF MENTAL HEALTH

“Taking care of your mental health is an act of self-love.”

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# Overview

What we'll discuss  
today

- **Goal -**

Create a tool to identify individuals who might need mental health support, so that healthcare professionals can offer help early.

- **Stakeholders -**

Psychologists, psychiatrists, general practitioners, and mental health counselors to prioritize interventions, allocate resources, and improve patient outcomes.

- **Business question –**

Who is likely to seek mental health treatment?

# Data Overview

- **Source** – Kaggle Mental Health Dataset

<https://www.kaggle.com/datasets/divaniazzahra/mental-health-dataset>

- **Size of the dataset** - 292809 rows, 17 columns
- **Target Variable** -Treatment
- **Features**



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 292364 entries, 0 to 292363
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Timestamp              292364 non-null  datetime64[ns]
1   Gender                 292364 non-null  object
2   Country                 292364 non-null  object
3   Occupation              292364 non-null  object
4   self_employed           287162 non-null  object
5   family_history           292364 non-null  object
6   treatment               292364 non-null  object
7   Days_Indoors            292364 non-null  object
8   Growing_Stress         292364 non-null  object
9   Changes_Habits          292364 non-null  object
10  Mental_Health_History    292364 non-null  object
11  Mood_Swings              292364 non-null  object
12  Coping_Struggles         292364 non-null  object
13  Work_Interest            292364 non-null  object
14  Social_Weakness          292364 non-null  object
15  mental_health_interview  292364 non-null  object
16  care_options             292364 non-null  object
dtypes: datetime64[ns](1), object(16)
memory usage: 37.9+ MB
```

# Project Workflow



# Importing Data and Preprocessing

1. Import data using Spark.
2. Convert Spark Dataframe to Pandas Dataframe.
3. Data Preprocessing
  - Handled missing values.
  - Removed duplicates.
  - Dropped Timestamp column.
  - Group rare categories in Country into ‘Other’.

```
Total number of rows before preprocessing: (292364, 17)
Total number of rows after preprocessing: (286808, 16)
```

```
Data Info:
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 286808 entries, 0 to 286807
```

```
Data columns (total 17 columns):
```

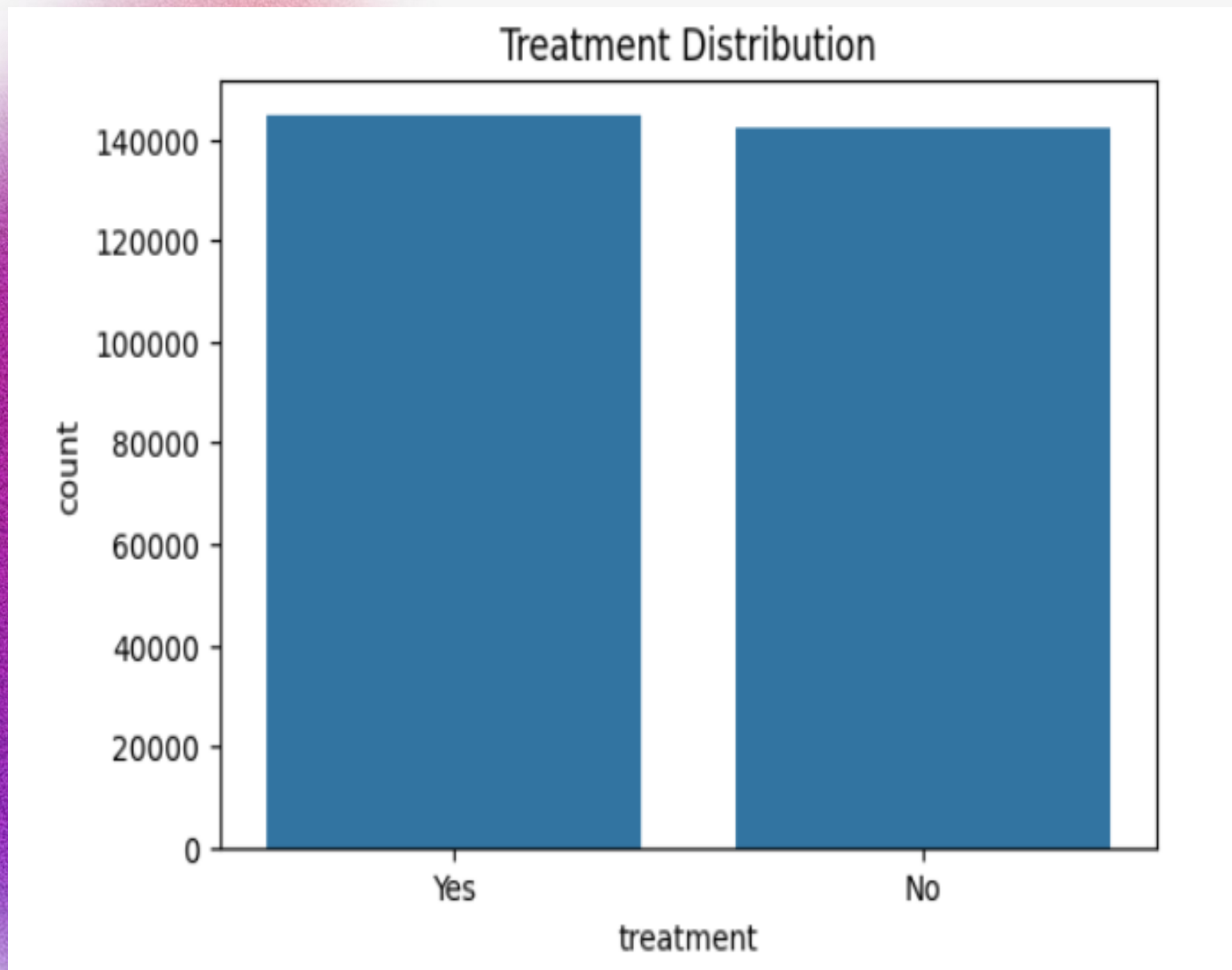
| #  | Column                  | Non-Null Count  | Dtype  |
|----|-------------------------|-----------------|--------|
| 0  | index                   | 286808 non-null | int64  |
| 1  | Gender                  | 286808 non-null | object |
| 2  | Country                 | 286808 non-null | object |
| 3  | Occupation              | 286808 non-null | object |
| 4  | self_employed           | 286808 non-null | object |
| 5  | family_history          | 286808 non-null | object |
| 6  | treatment               | 286808 non-null | object |
| 7  | Days_Indoors            | 286808 non-null | object |
| 8  | Growing_Stress          | 286808 non-null | object |
| 9  | Changes_Habits          | 286808 non-null | object |
| 10 | Mental_Health_History   | 286808 non-null | object |
| 11 | Mood_Swings             | 286808 non-null | object |
| 12 | Coping_Struggles        | 286808 non-null | object |
| 13 | Work_Interest           | 286808 non-null | object |
| 14 | Social_Weakness         | 286808 non-null | object |
| 15 | mental_health_interview | 286808 non-null | object |
| 16 | care_options            | 286808 non-null | object |

```
dtypes: int64(1), object(16)
```

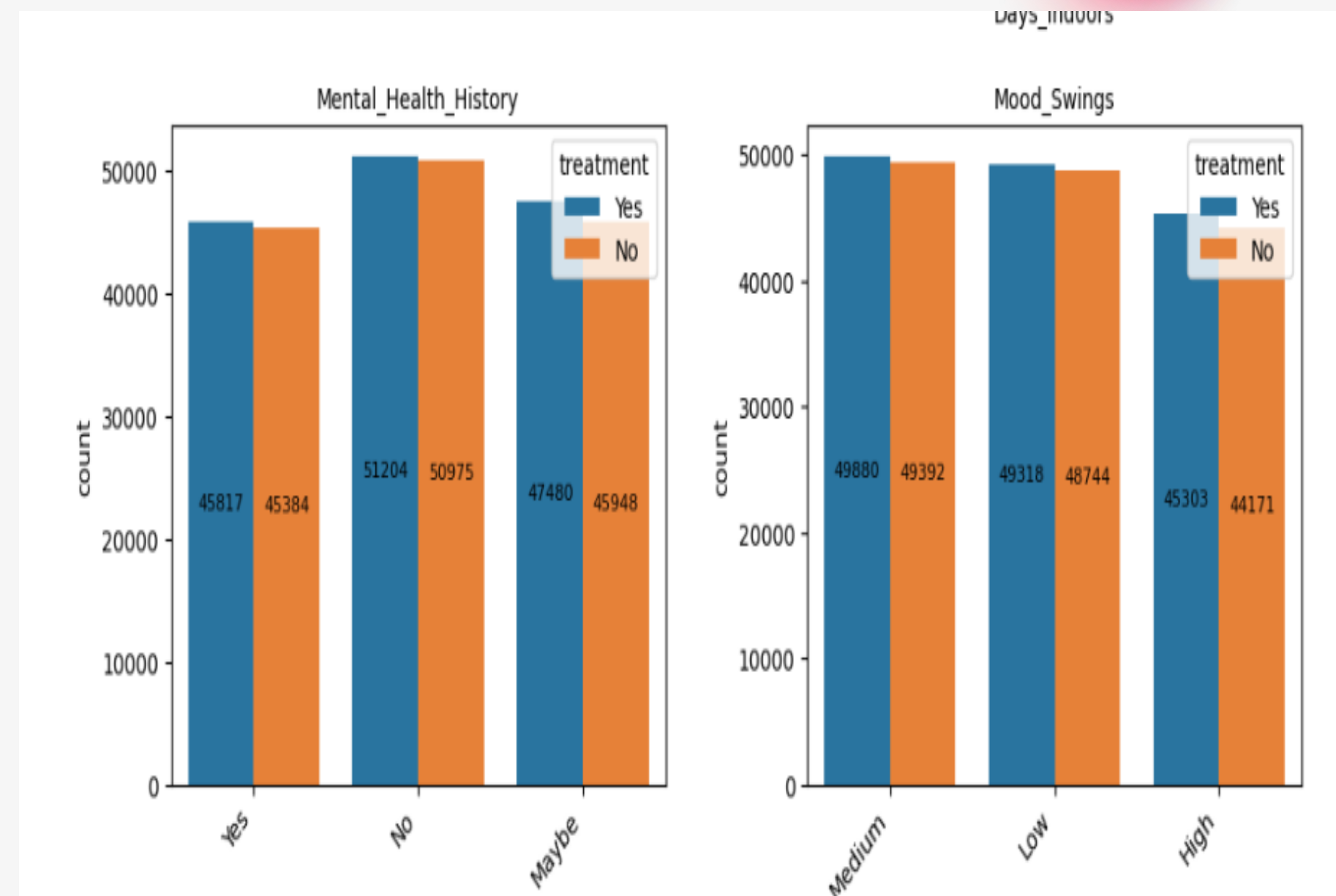
```
memory usage: 37.2+ MB
```

# Exploratory Analysis

## Target Distribution

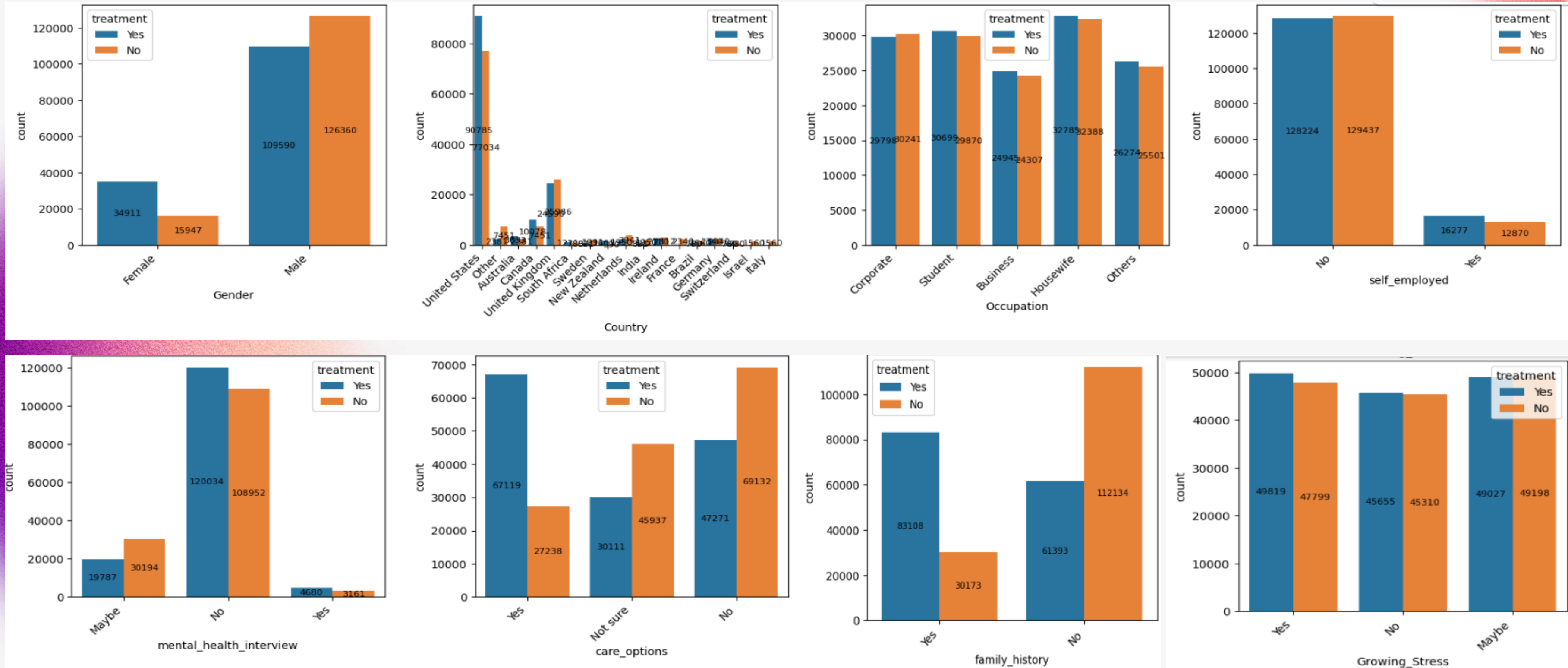


## Categorical Feature Analysis



# Exploratory Analysis

## Categorical Feature Analysis



# Splitting, Training and Evaluating

## Catboost model with 15 features

### Deliverable 1

```
CatBoost Model's performance with:  
iterations = 500  
depth = 6  
learning_rate = 0.1  
verbose = 0,  
random_state = 42,  
stopping_round = 50
```

```
-----  
Accuracy : 0.7860255918552351  
-----
```

Classification Report :

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| No           | 0.82      | 0.72   | 0.77     | 28252   |
| Yes          | 0.76      | 0.85   | 0.80     | 29110   |
| accuracy     |           |        | 0.79     | 57362   |
| macro avg    | 0.79      | 0.79   | 0.78     | 57362   |
| weighted avg | 0.79      | 0.79   | 0.79     | 57362   |

### Deliverable 2

```
CatBoost Model's performance with:  
iterations = 649  
depth = 8  
learning_rate = 0.05  
verbose = 0,  
random_state = 42,  
stopping_round = 100
```

```
-----  
Accuracy : 0.7859907255674489  
-----
```

Classification Report :

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| No           | 0.82      | 0.72   | 0.77     | 28252   |
| Yes          | 0.76      | 0.85   | 0.80     | 29110   |
| accuracy     |           |        | 0.79     | 57362   |
| macro avg    | 0.79      | 0.79   | 0.78     | 57362   |
| weighted avg | 0.79      | 0.79   | 0.79     | 57362   |

# Splitting, Training and Evaluating

## Logistic Regression model using 15 features

```
Training Data Score: 0.7091036670937824
Testing Data Score: 0.71306788466232
```

```
Prediction Results (First 10 Rows):
```

|   | Prediction | Actual |
|---|------------|--------|
| 0 | No         | No     |
| 1 | Yes        | No     |
| 2 | No         | Yes    |
| 3 | No         | No     |
| 4 | Yes        | No     |
| 5 | No         | No     |
| 6 | Yes        | No     |
| 7 | Yes        | Yes    |
| 8 | No         | No     |
| 9 | No         | No     |

```
Accuracy Score: 0.71306788466232
```

## Multi Layer Perceptron algorithm using 15 features

```
Training Data Score: 0.781970485430123
Testing Data Score: 0.7835500854224051
```

```
Prediction Results (First 10 Rows):
```

|   | Prediction | Actual |
|---|------------|--------|
| 0 | No         | No     |
| 1 | No         | No     |
| 2 | Yes        | Yes    |
| 3 | No         | No     |
| 4 | Yes        | No     |
| 5 | No         | No     |
| 6 | Yes        | No     |
| 7 | Yes        | Yes    |
| 8 | Yes        | No     |
| 9 | No         | No     |

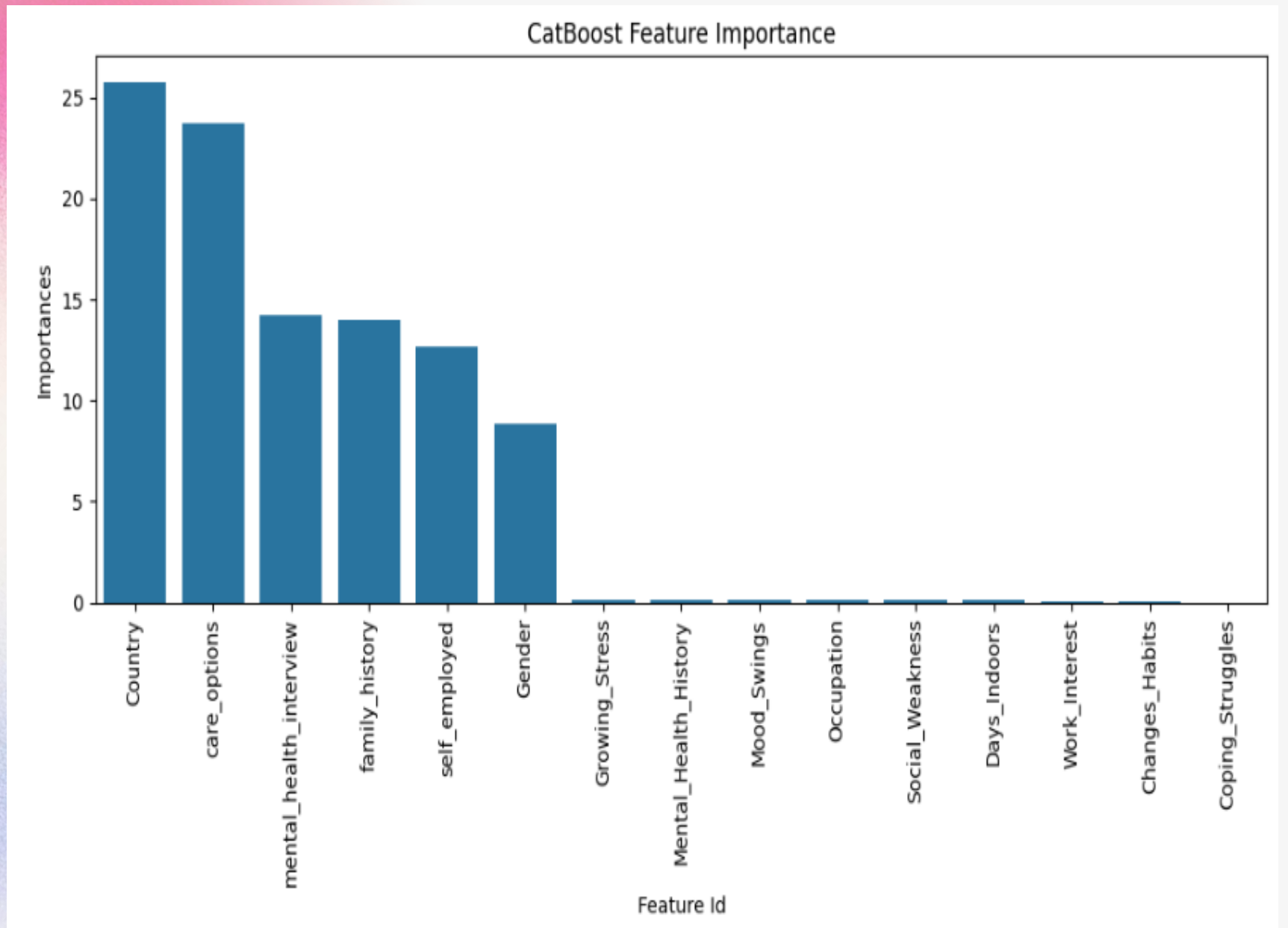
```
Accuracy Score: 0.7835500854224051
```

# Feature Selection

## Feature Importance on Catboost Model

Feature Importance :

|    | Feature Id              | Importances |
|----|-------------------------|-------------|
| 0  | Country                 | 25.765610   |
| 1  | care_options            | 23.762813   |
| 2  | mental_health_interview | 14.205505   |
| 3  | family_history          | 13.960962   |
| 4  | self_employed           | 12.679121   |
| 5  | Gender                  | 8.858460    |
| 6  | Growing_Stress          | 0.124753    |
| 7  | Mental_Health_History   | 0.114128    |
| 8  | Mood_Swings             | 0.109733    |
| 9  | Occupation              | 0.104349    |
| 10 | Social_Weakness         | 0.086439    |
| 11 | Days_Indoors            | 0.086400    |
| 12 | Work_Interest           | 0.077687    |
| 13 | Changes_Habits          | 0.060331    |
| 14 | Coping_Struggles        | 0.003709    |



# Correlation Analysis

- Measures how strongly features are related to each other.
- Measures how strongly each feature is related to the target variable.
- Drop highly correlated features to remove multicollinearity

Feature-Target Cramér's V Correlations:

Cramér's V with Treatment

|                         |          |
|-------------------------|----------|
| family_history          | 0.371379 |
| care_options            | 0.290716 |
| Country                 | 0.209506 |
| Gender                  | 0.169562 |
| mental_health_interview | 0.101948 |
| self_employed           | 0.036731 |
| Coping_Struggles        | 0.009680 |
| Growing_Stress          | 0.009635 |
| Occupation              | 0.008362 |
| Mental_Health_History   | 0.006167 |
| Days_Indoors            | 0.005206 |
| Work_Interest           | 0.003722 |
| Mood_Swings             | 0.003391 |
| Social_Weakness         | 0.002114 |
| Changes_Habits          | 0.001964 |

Highly Correlated Feature Pairs (Cramér's V  $\geq$  0.1):

[]

Features to Drop due to High Correlation: []

# Catboost Modeling using top 6 features

- The top 6 features selected are:
  1. Country
  2. care\_options
  3. mental\_health\_interview
  4. family\_history,
  5. self\_employed
  6. Gender

Deliverable 3

|                                                                                                                                                        |           |        |          |         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------|----------|---------|
| CatBoost Model's performance with:<br>iterations = 500<br>depth = 6<br>learning_rate =0.1<br>verbose = 0,<br>random_state = 42,<br>stopping_round = 50 |           |        |          |         |
| -----                                                                                                                                                  |           |        |          |         |
| Accuracy : 0.7859384261357694                                                                                                                          |           |        |          |         |
| -----                                                                                                                                                  |           |        |          |         |
| Classification Report :                                                                                                                                |           |        |          |         |
|                                                                                                                                                        | precision | recall | f1-score | support |
| No                                                                                                                                                     | 0.82      | 0.72   | 0.77     | 28252   |
| Yes                                                                                                                                                    | 0.76      | 0.85   | 0.80     | 29110   |
| accuracy                                                                                                                                               |           |        | 0.79     | 57362   |
| macro avg                                                                                                                                              | 0.79      | 0.78   | 0.78     | 57362   |
| weighted avg                                                                                                                                           | 0.79      | 0.79   | 0.78     | 57362   |

Deliverable 4

|                                                                                                                                                          |           |        |          |         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------|----------|---------|
| CatBoost Model's performance with:<br>iterations = 649<br>depth = 8<br>learning_rate =0.05<br>verbose = 0,<br>random_state = 42,<br>stopping_round = 100 |           |        |          |         |
| -----                                                                                                                                                    |           |        |          |         |
| Accuracy : 0.7852933998117221                                                                                                                            |           |        |          |         |
| -----                                                                                                                                                    |           |        |          |         |
| Classification Report :                                                                                                                                  |           |        |          |         |
|                                                                                                                                                          | precision | recall | f1-score | support |
| No                                                                                                                                                       | 0.81      | 0.73   | 0.77     | 28252   |
| Yes                                                                                                                                                      | 0.76      | 0.84   | 0.80     | 29110   |
| accuracy                                                                                                                                                 |           |        | 0.79     | 57362   |
| macro avg                                                                                                                                                | 0.79      | 0.78   | 0.78     | 57362   |
| weighted avg                                                                                                                                             | 0.79      | 0.79   | 0.78     | 57362   |
| -----                                                                                                                                                    |           |        |          |         |

# Catboost Modeling using top 10 features

- The top 6 features selected are:
  - Country
  - care\_options
  - mental\_health\_interview
  - family\_history,
  - self\_employed
  - Gender
  - Growing\_stress
  - mental\_health\_history
  - mood\_swing
  - occupation

Deliverable 5

```
CatBoost Model's performance with:
iterations = 649
depth = 8
learning_rate =0.05
verbose = 0,
random_state = 42,
stopping_round = 100

-----

Accuracy : 0.7859209929918761
-----

Classification Report :
              precision    recall  f1-score   support

      No         0.82         0.72         0.77        28252
      Yes         0.76         0.85         0.80        29110

 accuracy         0.79
 macro avg         0.79         0.79         0.78        57362
weighted avg         0.79         0.79         0.79        57362

-----

Model saved as ten_features_model_opt2.pkl
```

| FEATURES         | CATBOOST Accuracy | MLP Accuracy | LOGISTIC RERESSION Accuracy |
|------------------|-------------------|--------------|-----------------------------|
| 15 optimization1 | 78.60             | 78.35        | 71.30                       |
| optimization2    | 78.59             |              |                             |
| 6 optimization1  | 78.59             |              |                             |
| optimization2    | 78.52             |              |                             |
| 10 optimization1 | 78.59             |              |                             |
| optimization2    | 78.61             |              |                             |

Optimization 1 = iterations = 500, depth = 6, learning\_rate = 0.1  
Optimization 2 = iterations = 649, depth = 8, learning\_rate =0.05

# Model Chosen for Deployment

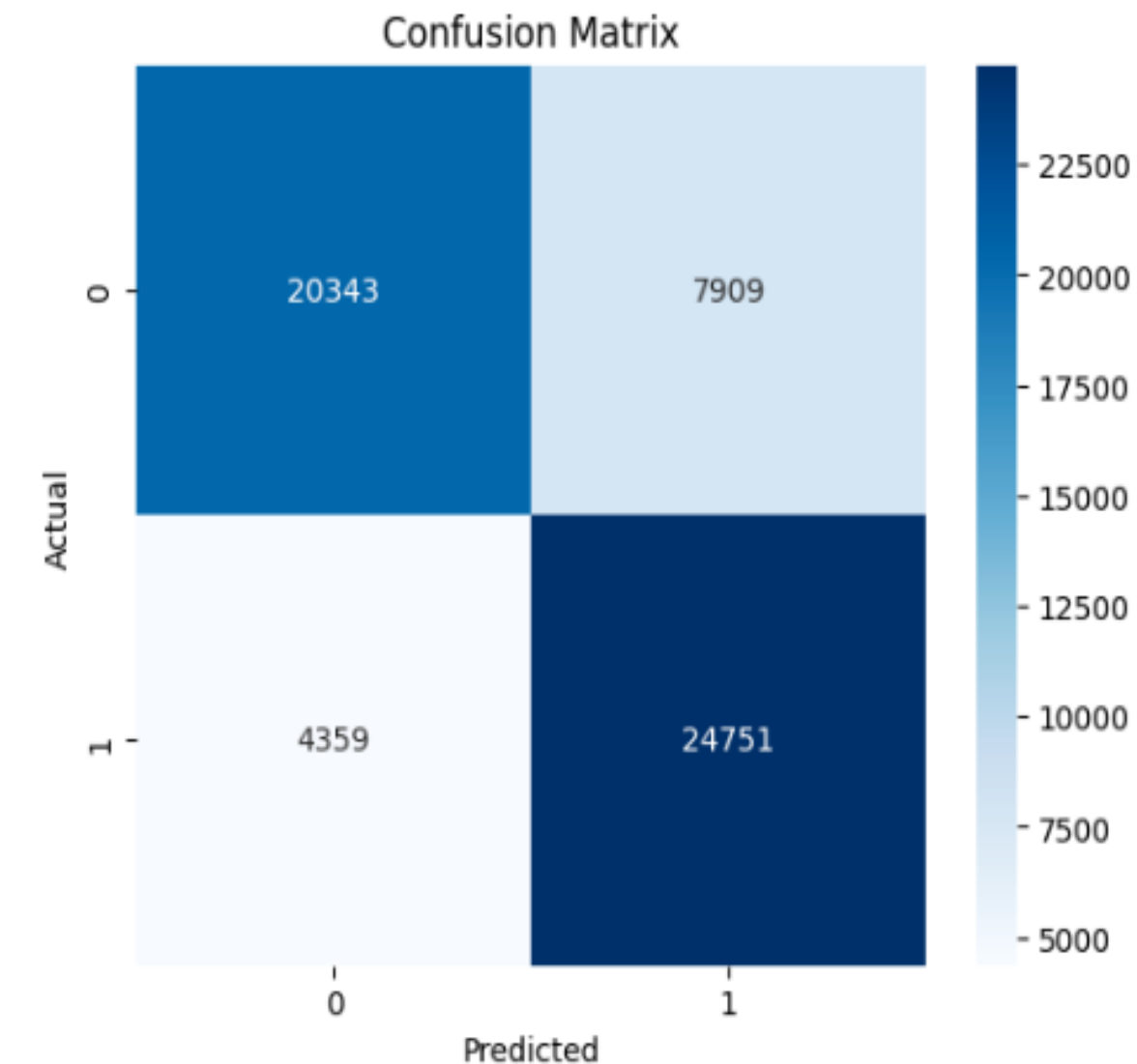
```
CatBoost Model's performance with:  
iterations = 500  
depth = 6  
learning_rate = 0.1  
verbose = 0,  
random_state = 42,  
stopping_round = 50
```

```
-----  
Accuracy : 0.7861301907185942  
-----
```

```
Classification Report :  
              precision    recall  f1-score   support  
  
   No           0.82         0.72         0.77       28252  
   Yes          0.76         0.85         0.80       29110  
  
 accuracy              0.79              0.79         0.79       57362  
 macro avg           0.79         0.79         0.78       57362  
weighted avg           0.79         0.79         0.79       57362
```

```
-----  
Model saved as ten_features_model_opt1.pkl  
-----  
-----
```

Confusion Matrix:



The model has been evaluated successfully and is ready for deployment in the cloud.

# Model Deployment

## **Flask Deployment:**

- The machine learning model was integrated into a Flask web application, enabling real-time predictions through a RESTful API. This setup ensured seamless client interactions and efficient model deployment.

## **AWS EC2 Hosting:**

- The Flask application was hosted on an AWS EC2 instance, providing scalability and reliable performance. This hosting solution supported robust management of computational resources for the deployed model.

# Future Improvements

- **Wearable Device Integration:**

Enhance the mental health prediction model by integrating real\_time health data from wearable devices like Fitbit or Apple Watch to improve accuracy and personalization of predictions.

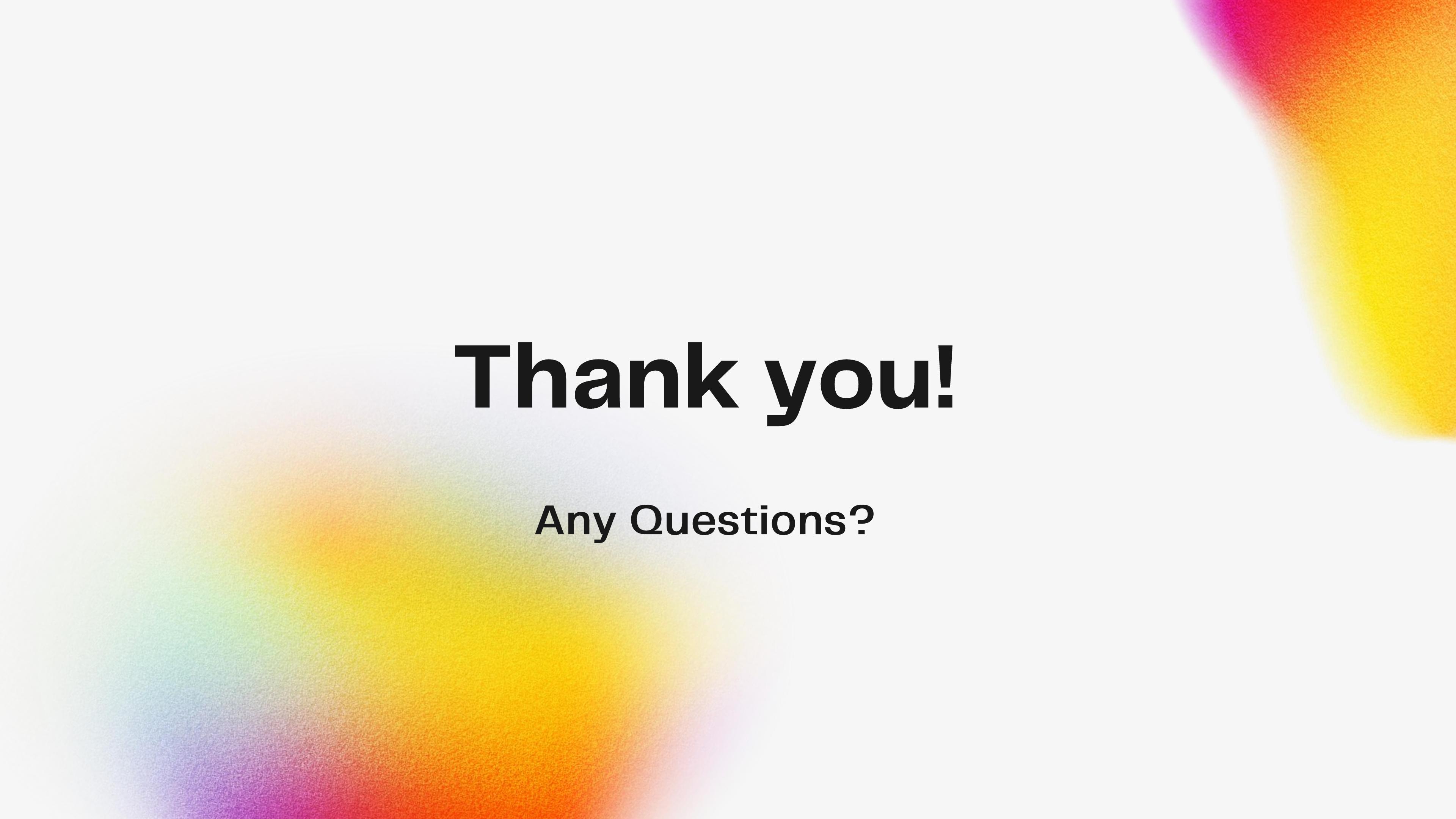
- **Real-Time Sentiment Analysis:**

Integrate natural language processing (NLP) to analyze user inputs (e.g., journal entries or social media activity) for real-time sentiment tracking, enhancing early detection of mental health risks.

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# Conclusion

- **Early Detection:** Identifies mental health risks early, enabling proactive interventions to improve outcomes.
- **Personalized Support:** Delivers tailored mental health recommendations based on individual factors like lifestyle and family history.
- **Model Optimization:** Enhanced through hyperparameter tuning for robust and meaningful predictions
- **Data-Driven Insights:** Achieves 78.6% accuracy using CatBoost algorithm.
- **Scalable Solution:** Deployed via a Flask application on AWS EC2, ensuring reliable, real-time insights for large populations.



# **Thank you!**

**Any Questions?**